

见微知著 - Decoding EEG Movement Imagination

Antonia Reul *University of Osnabrück*

Abstract—This article presents a CNN-Transformer for movement imagination decoding from EEG data. For imagined left or right fist motion, a hybrid CNN-Transformer architecture was chosen to benefit from the local feature extraction strength of CNNs and attentional mechanisms of a Transformer for global relationships. In order to reduce overfitting of the scarce EEG data, learning rate warm-up, swish and exponential linear units, and weight decay have been employed. The code of this project can be accessed in the accompanying GitHub repository (<https://github.com/Nia9-4/CNN-Transformer-EEG-BCI-26>).

I. INTRODUCTION

AFTER spinal cord injuries, patients may find themselves restricted in their motor skills due to an impaired communication between the muscle spindles and higher cognitive areas [14]. Brain Computer Interfaces (BCIs) can help patients regain motor abilities by enabling the control of external devices or systems based on the user's measured brain activity [35]. A promising application, which inspired this project, is an EEG-based BCI for limb movement based on movement imagination (MI), which can for instance be used for patients with phantom limbs. BCIs typically first acquire data from either noninvasive skull measurements, using electroencephalography (EEG), or invasive methods like electrocorticography (ECoG) [1]. The signals are then processed, relevant features extracted, classified, and the feedback is finally used to optimize the procedure [1]. This project simulates such process from EEG data to classification, and thereby aims to partially build on [23] and [21], but inspiration from further recent scientific papers and publications has been drawn for model improvement, as well.

A. Movement Imagination Decoding with Deep Learning

EEG signals are composed of different waves of distinct frequencies. For movement imagery, mu rhythms, waves oscillating between 8 and 12 Hz over motor and primary sensory cortical areas, are specifically of interest since those rhythms are suppressed during movement planning and execution [7]. Besides mu rhythms, beta signals (13 - 30 Hz) are also relevant in MI [26]. Different deep learning (DL) models have been applied to decode MI, among those being CNNs [20], RNNs [38], LSTMs [37], GRUs [5], and Transformers [40]. While CNNs excel at local feature extraction, they struggle with global patterns [2]. For Transformers the opposite is the case: Since they have no inductive local bias, they require large datasets to converge [2]. However, EEG data is usually highly variable, non-stationary, typically scarce, and has a low signal-to-noise ratio [21], and a Transformer may thereby quickly overfit the data. To tackle those challenges, research has focused on hybrid architectures, like CNN-Transformers, in recent years. Moreover, some BCI users fail to operate their

systems as intended, a phenomenon first described in 2010 as “BCI illiteracy” [4], which occurs in up to 10 - 50 % of BCI [13]. BCI illiteracy is another reason why the field is focusing more on hybrid Transformer-based architectures, which may help improve neural-signal separability [25].

II. METHODS

A. Dataset

The EEG Motor Movement/Imagery Dataset [29] is one of the standard BCI datasets besides the BCI Competition IV 2a dataset [34]. The dataset contains a set of 64-channel EEGs from participants performing motor/imagery (MI) tasks [29]. This project was interested in the experimental runs 4, 8 and 12, in which patients imagined either left or right fist movement. Subjects 88, 89, 92 and 100 have been excluded from the 109 participants for analysis (N = 105) since subject 89 contained incorrect labels and the other three subjects used a sampling rate of 128 Hz instead of 160 Hz [18], [28]. EEG data is usually three-dimensional and contains the number of trials (epochs), of EEG channels, and of time samples per epoch. Epochs themselves are two-dimensional arrays of shape (channels x time).

B. Data Preprocessing

As previously highlighted, EEG data is quite noisy and signals from the scalp are not necessarily accurately representing signals coming from the brain [11]. Eye blinks, muscle movements and other neural activity might distort the EEG signal, contaminating the data [15]. Therefore, pre-processing can help get closer to the “true” neural signal, before actually letting the CNN discover local, and the Transformer global dependencies. Pre-processing measures implemented in this project include band-pass and notch filtering, artifact removal via independent component analysis (ICA), epoching and normalization. Both the discriminative mu and beta event-related desynchronization (ERD) occur in the range of 40 Hz, as described above. Other MI related DL architectures use frequencies from 0 - 38 Hz [32], while the Filter Bank Common Spatial Pattern (FBCSP) ML pipeline for motor imagery traditionally selects frequencies in the range between 4 - 40 Hz [6]. The epoching window is chosen so that the first 500 ms of preparation latency before ERD onset and 600 ms before the 4.1 s task boundary, during which the beta-rebound transition usually happens, are excluded. The beta rebound is a very strong signal occurring post task, but including it may lead to the network interpreting it as task related. The number of temporal filters F1 has been chosen inspired by the CNN architecture ShallowNet [16]. For the code the MNE python library [24] has been used and inspiration drawn from [33].

C. CNN-Transformer

The EEG decoder should account for inter-subject variability and is therefore a cross subject model.

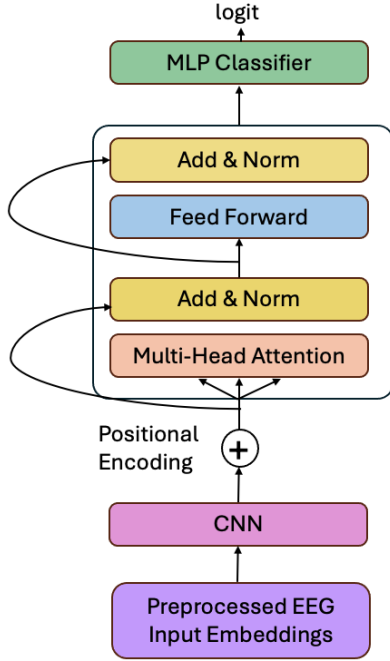


Fig. 1. The main model architecture.

1) *CNN*: When analyzing EEG data, temporal should occur before spatial filtering. If 1D convolutional layers were used, instead of 2D, the information from all electrodes would get mixed up. Instead, the goal should be to analyze the time dimension for each electrode independently, using a temporal filter first, as has been done in popular networks like EEGNet [20]. After every convolution, batch normalization is used to ensure stability since EEG signals are noisy and highly variable between subjects [31]. Only 16 temporal filters (F1) are used since they have been shown to generally be sufficient to cover primary frequency ranges of interest in MI [16], [20]. Using too many filters, like 64, may lead to the model drastically overfitting. Besides that, for every temporal filter D spatial filters are created, so that the model can look for different spatial topographies for the same frequency. Moreover, the pool size for the global average pooling has also been chosen in accordance with [16], [20]. One could also experiment with the values 8 and 2 instead, to compare how model performance improves or worsens. Further, ELU is used instead of ReLU since it allows for negative values which is useful when dealing with zero-mean oscillatory signals. For EEG data, the exponential linear unit (ELU) is almost always preferred over rectified linear units (ReLU) as activation function since ReLU clips and hence ignores negative values [31]. Because EEG data contains voltage fluctuations oscillating around a zero point, this would mean discarding crucial signal variance. ELU (1), in contrast, allows for negative values [8].

$$f(x) = \begin{cases} x & \text{if } x > 0 \\ \alpha(e^x - 1) & \text{if } x \leq 0 \end{cases} \quad (1)$$

2) *Encoder-Only Transformer*: Before Input can pass to the Transformer, the input tensor needs to be permuted since the Transformer expects the tensors to have the shape (batch, time samples, embeddings), and not (batch, embeddings, time samples). Because the goal is classification, and not sequence generation or translation of the data into another domain, an encoder-only architecture was chosen. The Transformer is built on a multi-head attention layer (2) (3) [36] and a feed-forward network, followed by a normalization layer and a sigmoid-weighted linear unit (SiLU) (??), also known as Swish activation function [?], [21].

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (2)$$

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O \quad (3)$$

While ELU is used in the CNN for reasons described above, SiLU, utilizing a "soft floor", allows for some values below zero [9] and has proven to perform very well for high-level feature processing [12], [27]. Since the Transformer receives the features extracted by the CNN and is expected to act like a global pattern-extractor, SiLU has been chosen.

$$f(x) = x \cdot \sigma(x) = \frac{x}{1 + e^{-x}} \quad (4)$$

To ensure stable learning during training and prevent vanishing gradients, residual connections are applied (??). Instead of applying post layer normalization, as has been done in [36], pre-LN is used to create a more stable gradient flow, which has reportedly been recommended for EEG-BCIs [21]

$$y = F(x) + x \quad (5)$$

3) *MLP Classifier*: The learned representations from the Transformer finally need to get mapped to the available classes. Logits are used here to not only highlight the predicted class, but also provide information about the classification confidence. The multilayer perceptron (MLP) classifier returns a logit for every sample in the batch - large positive values thereby signal a high confidence in class 1, while large negative values signal a high confidence for class 0. In MI, the brain usually produces sustained drops or increases in power, like Mu-rhythm desynchronization, over a few seconds [3]. To ignore short-duration noise, evoking single spikes, average pooling is used to observe how active the frequency band was throughout the trial. Sometimes concatenation is also used in MI classification tasks. Concatenation gives the MLP the decision between learning to ignore the maximum signal if it is too noisy, and to rely on the averaged part instead, or vice versa. As suggested by [20], a dropout rate of 0.5 is further used for the classifier.

D. Baseline Comparison Model

Employing convolutional and pooling layers, CNNs act as efficient feature extractors and serve as useful baseline models. However due to their local inductive bias, they struggle with capturing global relationships, making them unsuitable as standalone models for MI classification.

E. Training

In order to check for exploding or vanishing gradients and other unwanted phenomena during training, the dataset is split into a train, validation and test set with a 70% / 10% / 20% distribution. The input is first passed through the CNN, then the Transformer and finally classified by the MLP. For the backward pass, the loss is back-propagated through the network to calculate the gradients of the loss function with respect to the weights and biases, using binary cross entropy as the loss function (6) [19].

$$-(y \log(p) + (1 - y) \log(1 - p)) \quad (6)$$

The computed gradients from the backward pass are then used to update the network's parameters using the optimization algorithm Adam [?]. Weight decay, also known as L2 regularization, was added to the optimizer to prevent the cross-subject model from simply memorizing noise of training data. Since the EEG dataset has only a few thousand trials, while the Transformer has millions of parameters, not using L2 regularization (7) could quickly lead to extreme overfitting.

$$J(w) = L(y, \hat{y}) + \lambda \|w\|_2^2 \quad (7)$$

At the beginning of training, weights are randomly initialized, which can, however, in the case of a hybrid architecture, very quickly lead the model into suboptimal regions of the loss landscape. To prevent the Transformer from overpowering the CNN and allow the CNN to stabilize its feature extraction, and the Transformer to initialize its attention patterns, learning rate warm-up is used. First, a small learning rate is employed which is then gradually increased to the target initial learning rate. Cosine annealing is then used to make it diverge to zero again.

F. Evaluation

As evaluation metrics, accuracy, the amount of correctly classified samples, and Cohen's kappa are used. Cohen's kappa is especially useful for scenarios with imbalanced data distributions [21] and has been highlighted as a BCI standard [30]. To compare the classification results for different hyperparameter tunings, here the learning rate, confusion matrices have been computed and are shown in Figure 3. Loss curves and Kappa value over epochs are displayed in Figure 2.

III. RESULTS

While both training and validation loss seem to decrease for the baseline model, the validation loss quickly rises for the main model, as can be seen in Figure 2, which implies overfitting. Although decreasing the learning rate seems to improve model performance for the hybrid architecture (Fig. 2 and Fig. 3), and the confusion matrices may suggest somewhat reliable classification, the main model drastically overfits the data. Reasons for that are very likely to be the small size of the dataset, as well as the sparse nature of EEG signals.

IV. DISCUSSION

With only 105 subjects, the dataset chosen for this project was too small to train a Transformer model, let alone a hybrid architecture, reliably since large datasets are usually required to do so. While data augmentation may circumvent overfitting, Gao and Diniz [10] have also demonstrated that the EEG decoding architecture inductive bias might matter more than the raw data volume, illustrating the importance of a good architectural choice, instead.

Besides data augmentation, interesting extensions and ideas for this project for improvement may include using a Temporal Convolutional Network (TCN) as [21] for time-series modeling and capturing nuanced temporal dynamics. To handle data non-stationarity, multivariate empirical mode decomposition (MEMD) could also improve the model's results [21]. Further, [15] have illustrated the high potential of source localization techniques, like beam forcing, to compare accuracy scores. Another extension could be applying Grad-CAM or gate-weight analyses which links prediction to oscillatory bands for interpretation, as done by [10], who also suggest implementing zero-shot transfer learning. Besides that, spherical positional encoding, as suggested by [39], could represent another fruitful extension. In order to reliably validate the architecture, the model should further also be tested on other EEG datasets, like the BCI Competition IV 2a dataset [34].

V. CONCLUSION

The here suggested architecture overfits the EEG data and should therefore be further improved to reliably decode movement imagination. It seems to fail to "see the large in the small" (见微知著), relying too much on specific patterns and noise of the sparse EEG signals, instead of learning meaningful representations.

VI. ACKNOWLEDGMENTS

Laptops do not like tea. I am therefore very grateful for the support I received from Dr. Sam Jones and Bangor university when a liquid accident destroyed my laptop a few weeks/couple of days before the project submission deadline. Due to this accident, the final model is limited, does not perform as intended, and will therefore be further improved. I also want to thank Ulf Krumnack and Robin Rawiel for having offered this insightful course which inspired me to work on more DL projects in the future. Regarding the use of AI, the qwen3:8b and gemma4:31b-cloud model have been used for support with LaTeX formations, as well as during code debugging.

APPENDIX
FIGURE 2: LOSS CURVES

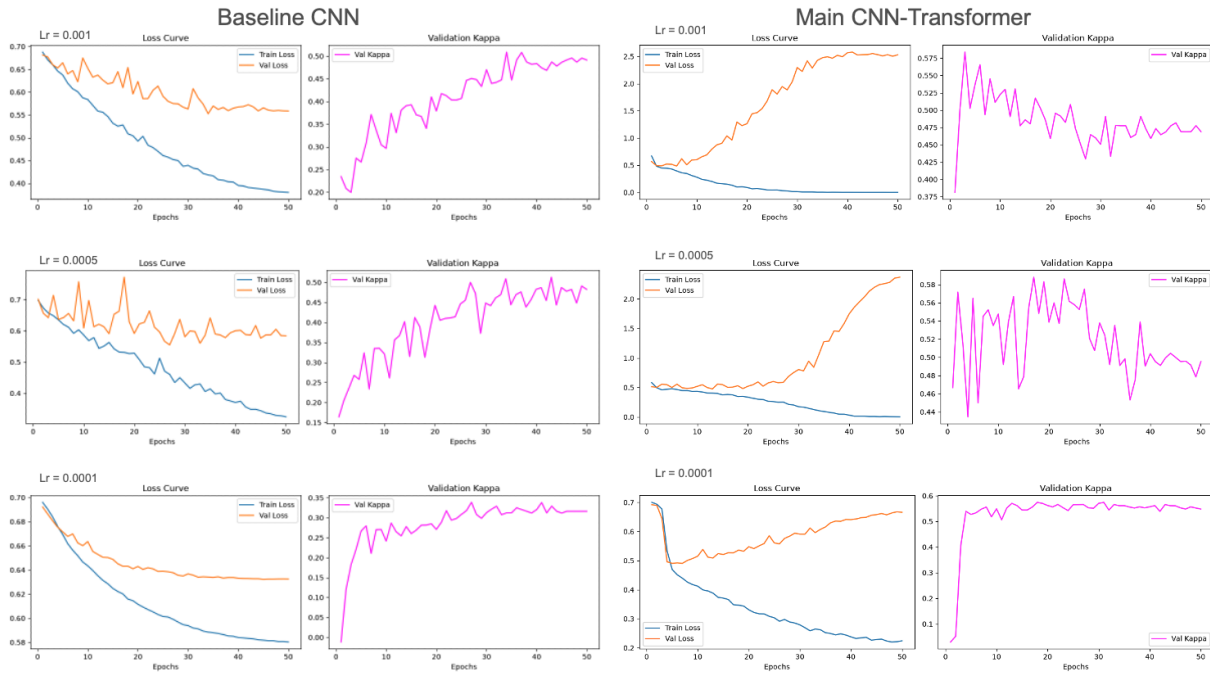


Fig. 2. Loss curves of the baseline and main model for different learning rates.

FIGURE 3: CONFUSION MATRICES

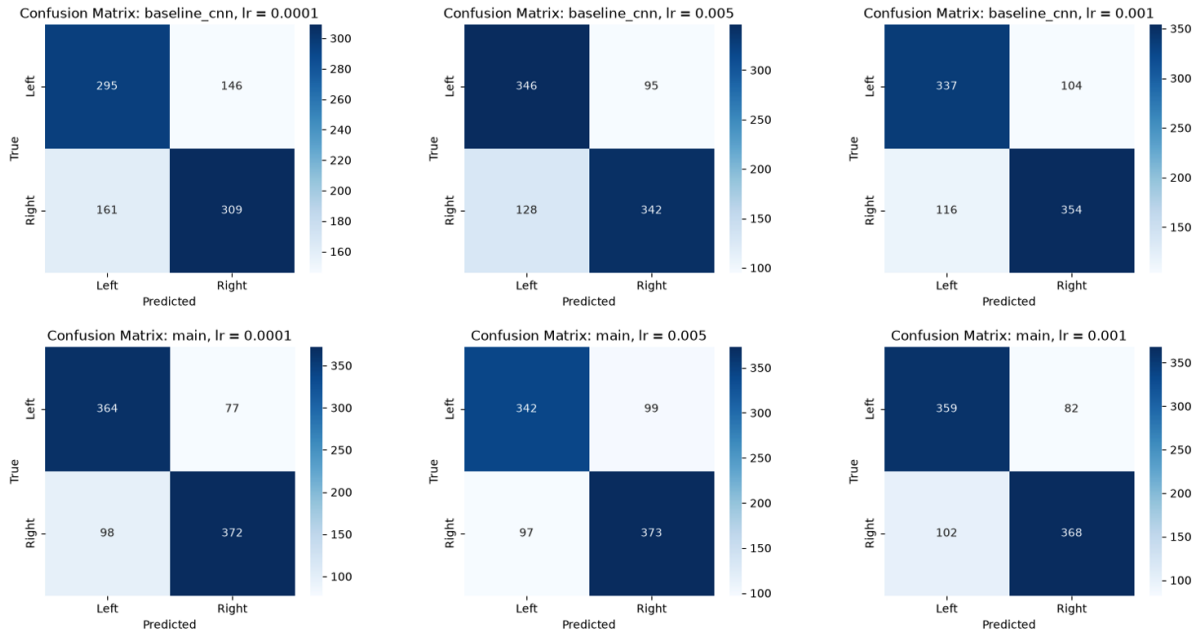


Fig. 3. Confusion matrices illustrating baseline and main model classification accuracy for different learning rates.

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